# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

**Full Syllabus (Chapters 1–13)** | Class 9 Science | Maximum Marks: 50

### MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** a simplified representation of a system that keeps what matters for the question | Models leave out unimportant details. |
| 2 | **(b)** a single kind of particle with fixed composition | Pure substances have fixed composition. |
| 3 | **(b)** Antoine Lavoisier | Lavoisier. |
| 4 | **(b)** electron | Cathode ray experiments. |
| 5 | **(c)** 0 | She returns to the starting point. |
| 6 | **(c)** Newton's first law | Law of inertia. |
| 7 | **(b)** a force acts on a body and the body is displaced in the direction of the force | W = F x s. |
| 8 | **(b)** vibrating bodies | Vibration is the source. |
| 9 | **(c)** Robert Hooke | In thin slices of cork, 1665. |
| 10 | **(c)** tissue | Tissue. |
| 11 | **(c)** biodiversity | Diversity of life. |
| 12 | **(b)** asexual reproduction | No gametes involved. |
| 13 | **(c)** biosphere | Sphere of life. |
| 14 | **(d)** an approximate answer is enough for a decision | Orders of magnitude. |
| 15 | **(d)** a heterogeneous mixture whose particles settle on standing | Large particles. |
| 16 | **(d)** 4 | Four atoms. |
| 17 | **(b)** protons | Z = p. |
| 18 | **(c)** acceleration | Change in velocity per time. |
| 19 | **(d)** are equal in magnitude, opposite in direction and act on different bodies | Third law. |
| 20 | **(c)** kinetic energy | W = change in KE. |
| 21 | **(c)** amplitude | Larger amplitude, louder. |
| 22 | **(d)** prokaryotic cell | e.g. bacteria. |
| 23 | **(a)** sclerenchyma | Dead, with lignin. |
| 24 | **(d)** algae | Thallophyta. |
| 25 | **(d)** stamen | Anther and filament. |
| 26 | **(c)** fresh snow | High albedo. |
| 27 | **(a)** a law describes what happens while a theory explains why | Different roles. |
| 28 | **(a)** pencil does not dissolve in the solvent | Ink would move. |
| 29 | **(d)** negatively charged ion | Gains electrons. |
| 30 | **(a)** 1 | Gains 1 to make 8. |
| 31 | **(b)** velocity, acceleration and displacement | Third equation. |
| 32 | **(b)** one-sixth | g_moon is g/6. |
| 33 | **(c)** makes work easier by changing the magnitude or direction of force | Machines help us. |
| 34 | **(a)** 17.2 m | v t/2 = 344 x 0.1/2. |
| 35 | **(d)** rough ER | Protein synthesis. |
| 36 | **(a)** epithelial tissue | Covering. |

| | | |
|---|---|---|
| **37** | **(c)** Porifera | Pores on the body. |
| **38** | **(b)** fruit | Fruit. |
| **39** | **(d)** stratosphere | About 15 to 35 km. |
| **40** | **(d)** a model | A visual model of matter. |
| **41** | **(a)** distilled water | Single compound. |
| **42** | **(d)** Joseph Proust | Proust. |
| **43** | **(b)** negative | Negative charge. |
| **44** | **(d)** greater than or equal to the magnitude of displacement | Path length is at least the straight-line gap. |
| **45** | **(d)** mass | More mass, more inertia. |
| **46** | **(d)** joule | 1 J = 1 N m. |
| **47** | **(a)** a vacuum | A material medium is required. |
| **48** | **(c)** Schleiden and Schwann | Plants and animals are made of cells. |
| **49** | **(b)** striated (voluntary) muscles | Skeletal. |
| **50** | **(d)** gills | Aquatic. |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERWING

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## ANSWER KEY & MARKING SCHEME SET 2

**Full Syllabus (Chapters 1–13)** | Class 9 Science | Maximum Marks: 50

### MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** testes | Male organ. |
| 2 | **(b)** water (hydrological) cycle | Cycle of water. |
| 3 | **(b)** 10000 litres | 0.5 x 20000. |
| 4 | **(b)** fog | Water droplets in air. |
| 5 | **(d)** 8 | Eight atoms. |
| 6 | **(b)** protons plus neutrons | A = p + n. |
| 7 | **(d)** displacement | Product of v and t. |
| 8 | **(c)** act on different bodies | Different bodies. |
| 9 | **(a)** can neither be created nor destroyed but only changed from one form to another | Law. |
| 10 | **(c)** frequency | Higher frequency, higher pitch. |
| 11 | **(b)** bacterium | No nuclear membrane. |
| 12 | **(b)** meristematic tissue | Dividing cells. |
| 13 | **(a)** taxonomy | Naming and grouping. |
| 14 | **(b)** binary fission | Splits into two. |
| 15 | **(a)** geosphere | Rocky part. |
| 16 | **(b)** a general principle that applies in every process | A fundamental law. |
| 17 | **(a)** moves farther up the paper | Greater solubility. |
| 18 | **(a)** $NaCl$ | 1:1. |
| 19 | **(d)** chemically unreactive | Stable. |
| 20 | **(c)** 10 m/s squared | g about 9.8. |
| 21 | **(c)** increasing the stopping time and reducing the force | F = change in momentum/time. |
| 22 | **(d)** load / effort | MA. |
| 23 | **(d)** echolocation using ultrasonic waves | Reflected ultrasound. |
| 24 | **(a)** lipid synthesis and detoxification | Fats and detox. |
| 25 | **(d)** xylem | Tracheids and vessels. |
| 26 | **(c)** bryophytes | Need water for fertilisation. |
| 27 | **(b)** carpel (pistil) | Stigma, style, ovary. |
| 28 | **(a)** ocean water or forests | Low albedo. |
| 29 | **(d)** describes a regular pattern observed in nature | Laws describe regularities. |
| 30 | **(b)** brass | Alloy has uniform composition. |
| 31 | **(c)** equals the total mass of the products | Mass is neither created nor destroyed. |
| 32 | **(d)** Rutherford | Gold foil experiment. |
| 33 | **(b)** vector quantity | It has magnitude and direction. |
| 34 | **(a)** newton | N = kg m/s squared. |
| 35 | **(d)** zero | Force is perpendicular to displacement. |
| 36 | **(c)** longitudinal waves | Particles vibrate along the direction of travel. |
| 37 | **(d)** new cells arise from pre-existing cells | All cells come from cells. |
| 38 | **(d)** squamous epithelium | Flat cells. |
| 39 | **(a)** Cnidaria (Coelenterata) | Stinging cells. |

| | | |
|---|---|---|
| **40** | **(d)** fertilisation | Zygote formation. |
| **41** | **(d)** harmful ultraviolet radiation | Absorbs UV. |
| **42** | **(d)** refine the model | Science corrects itself. |
| **43** | **(b)** smaller than the pores of the paper | Colloidal particles are small. |
| **44** | **(a)** 18 u | 2 + 16. |
| **45** | **(a)** 23 | 11 + 12. |
| **46** | **(a)** at rest | Distance does not change. |
| **47** | **(b)** the exhaust gases pushed downward exert an equal and opposite force on it | Third law. |
| **48** | **(b)** rate of doing work | P = W/t. |
| **49** | **(a)** distinguish two sounds of the same pitch and loudness | Waveform differs. |
| **50** | **(a)** mitochondrion | Releases energy. |

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## ANSWER KEY & MARKING SCHEME SET 3

**Full Syllabus (Chapters 1–13)** | Class 9 Science | Maximum Marks: 50

### MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** smooth (involuntary) muscles | Unstriated. |
| 2 | **(b)** Amphibia | Live on land and water. |
| 3 | **(b)** ovaries | Female organ. |
| 4 | **(c)** cycling of nutrients through living and non-living parts of the Earth | Matter recycles. |
| 5 | **(a)** burning of magnesium | New substance formed. |
| 6 | **(c)** homogeneous mixtures of metals | Metal solutions. |
| 7 | **(d)** $CaCl_2$ | Ca2+ and Cl-. |
| 8 | **(b)** atomic number but different mass numbers | Same Z. |
| 9 | **(b)** 20 m/s | v = gt. |
| 10 | **(c)** increase the time of stopping and reduce the force | Impulse. |
| 11 | **(c)** fulcrum | Pivot. |
| 12 | **(a)** sound navigation and ranging | Uses ultrasonic waves. |
| 13 | **(a)** DNA and proteins | Hereditary material. |
| 14 | **(c)** intercalary meristem | At the base of leaves/internodes. |
| 15 | **(c)** Carolus Linnaeus | Two-name system. |
| 16 | **(b)** budding | Buds from parent. |
| 17 | **(a)** hydrosphere | Water. |
| 18 | **(b)** explains many observations and makes testable predictions | Theories explain and predict. |
| 19 | **(d)** heterogeneous | Components are visible. |
| 20 | **(b)** 44 g | 12 + 32 = 44. |
| 21 | **(b)** James Chadwick | 1932. |
| 22 | **(b)** scalar quantity | Only magnitude. |
| 23 | **(b)** 1 kg | F = ma. |
| 24 | **(a)** 50 J | F s. |
| 25 | **(a)** compression | High pressure region. |
| 26 | **(a)** unicellular | e.g. Amoeba. |
| 27 | **(d)** phloem | Sieve tubes. |
| 28 | **(d)** pteridophytes | Ferns. |
| 29 | **(a)** pollen grains | Male gametes. |
| 30 | **(c)** sunlight falls obliquely and is spread over a larger area | Less heating. |
| 31 | **(d)** units must be used carefully and consistently | Importance of units. |
| 32 | **(c)** centrifugation | Spinning. |
| 33 | **(b)** 44 u | 12 + 32. |
| 34 | **(c)** 18 | 35 - 17. |
| 35 | **(a)** uniform velocity | Velocity constant. |
| 36 | **(c)** opposite to the direction of motion or tendency of motion | Opposes relative motion. |
| 37 | **(c)** watt | 1 W = 1 J/s. |
| 38 | **(d)** 20 Hz to 20000 Hz | Normal hearing. |

| | | |
|---|---|---|
| **39** | **(d)** Golgi apparatus | Golgi. |
| **40** | **(a)** columnar | Absorption. |
| **41** | **(c)** Platyhelminthes | Flat body. |
| **42** | **(b)** zygote | Fertilised egg. |
| **43** | **(b)** chlorofluorocarbons (CFCs) | Used in refrigerators and sprays. |
| **44** | **(d)** new evidence required changes | Models are revised. |
| **45** | **(a)** filtration | Insoluble solid. |
| **46** | **(b)** $Al_2O_3$ | Al3+ and O2-. |
| **47** | **(c)** mass number but different atomic numbers | Same A. |
| **48** | **(a)** zero | Instantaneously at rest. |
| **49** | **(a)** stop passengers from moving forward due to inertia | Inertia of motion. |
| **50** | **(c)** between the load and the effort | e.g. scissors, seesaw. |

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